

1. Perform the following matrix-vector multiplications:

c.
$$\begin{bmatrix} 2 & 0 & 0 \\ 3 & -1 & 2 \\ 0 & 2 & -3 \end{bmatrix} \begin{bmatrix} 2 \\ 5 \\ 1 \end{bmatrix} = \begin{bmatrix} 2 \times 2 + 0 \times 5 + 0 \times 1 \\ 3 \times 2 + (-1) \times 5 + 2 \times 1 \\ 0 \times 2 + 2 \times 5 + (-3) \times 1 \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 7 \end{bmatrix}$$

5. Determine which of the following matrices are nonsingular and compute the inverse of those matrices:

a.
$$\begin{bmatrix} 4 & 2 & 6 \\ 3 & 0 & 7 \\ -2 & -1 & -3 \end{bmatrix}$$

b.
$$\begin{bmatrix} 1 & 2 & 0 \\ 2 & 1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$$

a)
$$\begin{array}{ccc|cc} 4 & 2 & 6 & 4 & 2 \\ 3 & 0 & 7 & 3 & 0 \\ -2 & -1 & -3 & -2 & -1 \end{array}$$

$$\begin{array}{ccccc} - & - & - & + & + & + \end{array}$$

$$\det(A) = (4)(0)(-3) + (2)(7)(-2) + (6)(3)(-1) - (6)(0)(-2) - (4)(7)(-1) - (2)(3)(-3)$$

$$= 0 - 28 - 18 - 0 + 28 + 18$$

$$= 0$$

$$\det = 0 \text{ so doesn't have an inverse}$$

b)
$$\begin{array}{ccc|cc} 1 & 2 & 0 & 1 & 2 \\ 2 & 1 & -1 & 2 & 1 \\ 3 & 1 & 1 & 3 & 1 \end{array}$$

$$\begin{array}{ccccc} - & - & - & + & + & + \end{array}$$

$$\det(B) = (1)(1)(1) + (2)(-1)(3) + (0)(2)(1) - (0)(1)(3) - (1)(-1)(1) - (2)(2)(1)$$

$$= 1 - 6 + 0 + 0 + 1 - 4$$

$$= -8$$

$$\det \neq 0 \text{ so has an inverse}$$

Determine whether the matrix has a determinant, if it does find the inverse.

$$A = \begin{bmatrix} 1 & 4 \\ 7 & 2 \end{bmatrix}$$

$$\det A = 1(2) - 4(7) = 2 - 28 = -26 \neq 0 \text{ so has an inverse}$$

$$A^{-1} = \frac{1}{\det} \begin{bmatrix} 2 & -4 \\ -7 & 1 \end{bmatrix} = \frac{1}{-26} \begin{bmatrix} 2 & -4 \\ -7 & 1 \end{bmatrix} = \begin{bmatrix} \frac{2}{-26} & \frac{-4}{-26} \\ \frac{-7}{-26} & \frac{1}{-26} \end{bmatrix} = \begin{bmatrix} -\frac{1}{13} & \frac{2}{13} \\ \frac{7}{26} & -\frac{1}{26} \end{bmatrix}$$